Representation Theory 2

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Remark 0.1. *For any ring R, we have a canonical isomorphism,*

$$R \underset{\mathbb{Z}}{\otimes} \Lambda_n \cong R \underset{\mathbb{Z}}{\otimes} \mathbb{Z}[T_1, \cdots, T_n]^{S_n} \cong R[T_1, \cdots, T_n]^{S_n}.$$

That is m_λ where λ is a partition at most n-parts form a basis of $R[T_1, \cdots, T_n]^{S_n}$.

Theorem 0.1. *The algebra Λ_n is generated by the elementary symmetric polynomials $e_1, \cdots, e_n$. Moreover, they are algebraically independent hence*

$$\mathbb{Z}[T_1, \cdots, T_n]^{S_n} \cong \mathbb{Z}[e_1, \cdots, e_n].$$

Proof. We prove it by induction on n. To avoid confusion, we will denote,

$$e_i^{(n)} := e_i \in \Lambda_n,$$

a i-th elementary symmetric polynomial in n variables. For $n = 1$, this is trivial. Suppose we have the statement for $n-1$ variables. Consider the following map,

$$\Lambda_n \ni f(T_1, \cdots, T_n) \overset{\pi_n}{\to} f(T_1, \cdots, T_{n-1}, 0) \in \Lambda_{n-1}.$$

Under this morphism, we have,

$$e_i^{(n)} \mapsto \begin{cases} e_i^{(n-1)}, & (i < n), \\ 0, & (i = n). \end{cases}$$

By the induction hypothesis, for any $f \in \Lambda_n$, there is $p \in \Lambda_{n-1}$ such that

$$\pi_n(f) = p(e_1^{(n-1)}, \cdots, e_{n-1}^{(n-1)}).$$

Then,

$$\pi_n(f - p(e_1^{(n)}, \cdots, e_n^{(n)})) = \pi_n(f) - p(e_1^{(n-1)}, \cdots, e_{n-1}^{(n-1)}) = 0.$$

Since the kernel of π_n is (T_n), we conclude $T_n | f - p(e_1^{(n)}, \cdots, e_n^{(n)})$. By symmetry of the argument, this holds for any T_i. Thus,

$$e_n^{(n)} | f - p(e_1^{(n)}, \cdots, e_n^{(n)}).$$

From this, set $f' \in \Lambda_n$ to be such that $(f', e_n^{(n)}) = 1$ and

$$f - p(e_1^{(n)}, \cdots, e_n^{(n)}) = f'(e_n^{(n)})^k.$$

Apply the same argument to f', we see

$$f' = p(e_1^{(n)}, \cdots, e_n^{(n)}).$$

For the algebraic independence, suppose we have,

$$p(e_1^{(n)}, \cdots, e_n^{(n)}) = 0,$$

for some $p \in \mathbb{Z}[e_1^{(n)}, \cdots, e_n^{(n)}]$ which is of minimal degree in T_n. Then $p \in \ker \pi_n$. Thus we have,

$$p = q_0 + q_1 T_n + q_2 T_n^2 + \cdots,$$

Note that $q_0(e_1^{(n-1)}, \cdots, e_{n-1}^{(n-1)}) = 0$. Iterate this step, we conclude

$$p = T_n^k \tilde{p}.$$

As Λ_n is a domain $\tilde{p}(e_1^{(n)}, \cdots, e_n^{(n)}) = 0$ but this contradicts to the minimality of p. □

Theorem 0.2. *Recall Theorem* **??** *and under the setting of it, the induced injection is an isomorphism,*

$$\gamma : \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n} \xrightarrow{\sim} \mathcal{O}(T)^{S_n}.$$

Proof. As the injectiity is already proven, we only need to show the surjectivity. Observe,

$$\mathcal{O}(T)^{S_n} = \mathbb{C}[T_1^{\pm 1}, \cdots, T_n^{\pm 1}]^{S_n} = \mathbb{C}[T_1, \cdots, T_n, (T_1 \cdots T_n)^{-1}].$$

Sufficient to show that $e_1, \cdots, e_n, (T_1 \cdots T_n)^{-1} \in \operatorname{im} \gamma$. As $\det^{-1}$ is invariant under the conjugation, we have

$$\overset{-1}{\det} \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}.$$

Clearly, we have,

$$\overset{-1}{\gamma(\det)} \begin{pmatrix} t_1 & & \\ & \ddots & \\ & & t_n \end{pmatrix} = (t_1 \cdots t_n)^{-1}.$$

Thus $\gamma(\det^{-1}) = (T_1 \cdots T_n)^{-1}$. Let us denote the characteristic polynomial of $A \in \mathrm{GL}_n$ in the variable T by,

$$p(A; T) = \det(I_n - TA) = I_n + a_1(A)T + \cdots + a_{n-1}(A)T^{n-1} + a_n(A)T^n.$$

Note that $a_1, \cdots, a_n \in \mathcal{O}(\mathrm{GL}_n)^{\mathrm{GL}_n}$ since p is invariant of conjugation. Substituting a diagonal matrix into A, we get,

$$\begin{aligned} p(\mathrm{diag}(t_1, \cdots, t_n); T) &= \det(\mathrm{diag}(1 - t_1 T, \cdots, 1 - t_n T)), \\ &= \prod_{i=1}^{n} (1 - t_i T), \\ &= 1 + \sum_{i=1}^{n} (-1)^n e_i(t_1, \cdots, t_n) T^i. \end{aligned}$$

Thus $\gamma(a_i) = e_i$. We conclude γ is surjective. □

Definition 0.1. *A polynomial $f \in \mathbb{Z}[T_1, \cdots, T_n]$ is alternating if for any $\sigma \in S_n$, we have,*

$$f(T_{\sigma(1)}, \cdots, T_{\sigma(n)}) = \operatorname{sgn}\sigma f(T_1, \cdots, T_n).$$

Definition 0.2. *The Vandermonde determinant is*

$$V(T_1, \cdots, T_n) = \prod_{i<j}(T_i - T_j).$$

Clearly this is alternating.

Proposition 0.1. *Any alternating polynomial is of the form gV where g is a symmetric polynomial. That is to say the space of alternating polynomial is a free Λ_n-module of rank* 1.

Proof. Let

$$f = \sum_{\alpha\in\mathbb{N}^n} c_\alpha \underline{T}^\alpha \in \mathbb{Z}[T_1, \cdots, T_n],$$

be alternating. Then for any $(a_1, \cdots, a_n)$, we have,

$$c_{(a_1,\cdots,a_i\cdots,a_j,\cdots,a_n)} = -c_{(a_1,\cdots,a_j\cdots,a_i,\cdots,a_n)}.$$

Also note that,

$$(T_i - T_j) \mid T_1^{a_1}\cdots T_i^{a_i}\cdots T_j^{a_j}\cdots T_n^{a_n} - T_1^{a_1}\cdots T_i^{a_j}\cdots T_j^{a_i}\cdots T_n^{a_n}.$$

Therefore

$$\forall 1 \le i < j \le n, f \equiv 0 \mod (T_i - T_j) \Rightarrow V \mid f.$$

Obviously $\frac{f}{V}$ is symmetric. The freeness follows from that $\mathbb{Z}[T_1, \cdots, T_n]$ being a domain. □

Remark 0.2. *The module of alternating polynomials is isomorphic to Λ_n as Λ_n-modules.*

Notation 0.1. *For a descending chain $m_1 > m_2 > \cdots m_n > 0$, we define,*

$$A_{m_1>\cdots>m_n} = \sum_{\sigma\in S_n} \operatorname{sgn}\sigma \prod_{i=1}^{n} T_{\sigma(i)}^{m_i}.$$

Clearly, this is alternating.

Lemma 0.1. *$A_{m_1>\cdots>m_n}$-s form a basis of the space of alternating functions.*

Proof. Let

$$f = \sum_{\alpha\in\mathbb{N}^n} c_\alpha \underline{T}^\alpha \in \mathbb{Z}[T_1, \cdots, T_n],$$

be alternating. By the argument in Proposition 0.1, we have,

$$c_\alpha \neq 0 \Rightarrow \exists\sigma \in S_n \,\text{s.t.}\, \alpha_{\sigma(1)} > \cdots > \alpha_{\sigma(n)} > 0.$$

As σ is generated by transposition, we conclude,

$$f = \sum_{m_1>\cdots>m_n>0} d_{m_1>\cdots>m_n} A_{m_1>\cdots>m_n}.$$

Explicitly,

$$d_{m_1>\cdots>m_n} = c_{(m_1,\cdots,m_n)}.$$

□

Lemma 0.2. *Denote $\rho = (n-1, n-2, \cdots, 2, 1, 0) \in \mathbb{Z}^n_{\geq 0}$. Then we have a bijection between*

$$\{\textit{partitions with at most n-parts}\} \underset{m-\rho \hookleftarrow m}{\overset{\lambda \mapsto \lambda+\rho}{\rightleftarrows}} \{\textit{strictly decreasing sequences of length } n\}$$

Definition 0.3. *For any partition λ with at most n-parts, we define,*

$$s_\lambda := \frac{A_{\lambda+\rho}}{V} \in \Lambda_n.$$

We call polynomials of this form Schur polynomials.

Corollary 0.1. *Schur polynomials form a basis of Λ_n.*

Proof. By Remark 0.2 and Lemma 0.1, a basis element $\{A_{m_1>\cdots>m_n}\}$ is mapped to $\{s_\lambda\}$. By Lemma 0.2, this induces a bijection. □

Definition 0.4. *Let λ, μ be partitions. The lexicographical order is defined as follows. $\lambda < \mu$ if there exist $N \in \mathbb{N}$ such that*

$$\forall i < N, \lambda_i = \mu_i, \lambda_N < \mu_N.$$

Proposition 0.2. *Motivated by Remark* **??** *and Corollary 0.1, we can consider the following base change,*

$$m_\lambda = \sum_{\substack{\mu \\ \textit{with at most} \\ \textit{n-parts}}} p_{\lambda,\mu} s_\mu.$$

The matrix $P = (p_{\lambda,\mu})_{\substack{\lambda,\mu \\ \textit{with at most} \\ \textit{n-parts}}}$ is lower unitriangular (i.e. the diagonals are all 1) with respect to the lexicographical order.

Proof. For any $f \in \mathbb{Z}[T_1, \cdots, T_n]$ we set $l(f)$ to be the leading monomial of f (i.e. the maximal monomial in the lexicographical order with non-zero coefficient). Then we have,

$$l(fg) = l(f)l(g).$$

Apply this to $A_{\mu+\rho}$, we get,

$$\begin{aligned} T_1^{\mu_1+n-1}\cdots T_n^{\mu_n} &= l(A_{\mu+\rho}),\\ &= l(s_\mu V),\\ &= l(s_\mu)l(V) = l(s_\mu)(T_1^{n-1}\cdots T_{n-1}^1 T_n^0).\\ &\Rightarrow l(s_\mu) = T_1^{\mu_1}\cdots T_n^{\mu_n}. \end{aligned}$$

Thus m_μ, s_μ have the same leading monomials. □

Remark 0.3. *Let P be the base change matrix, then $Q = P^{-1}$ is also lower unitriangular. $Q_{\lambda,\mu}$ is called a Kostka number.*

Lemma 0.3. *Consider,*

$$\mathbb{Z}[\![x_1,\cdots,x_n,y_1,\cdots,y_n]\!]^{\mathfrak{S}_n\times\mathfrak{S}_n}.$$

We have the following identity,

$$\prod_{1\le i,j\le n}\frac{1}{1-x_iy_j} = \sum_{\substack{\lambda,\mu\\ \text{with at most}\\ n\text{-parts}}} s_\lambda(x_1,\cdots,x_n)s_\mu(y_1,\cdots,y_n).$$

0.1 Characters for the Symmetric Groups

Definition 0.5. *Let $\lambda \vdash r$. A Young tableau of shape λ is a filling of the Young diagram of λ in such a way that each column and row are increasingly numbered by numbers from 1 to r.*

A Young tableau is superstandard if numbers are filled starting from the top left corner to the right consecutively.

Example 0.1. *For $\lambda = (4,4,2,1)$, a Young tableau of shape λ is*

1	2	4	6
3	5	8	11
7	9		
10			

The superstandard Young tableau of shape λ is

1	2	3	4
5	6	7	8
9	10		
11			

Definition 0.6. *Fix a Young tableau T of shape $\lambda \vdash r$.*

1). The row stabilizers is the subgroup $\mathfrak{R}_T$ of $\mathfrak{S}_r$ such that for any $\sigma \in \mathfrak{R}_T$ and $i = 1,\cdots,r$, $\sigma(i)$ is in the same row.

2). The column stabilizers is the subgroup $\mathfrak{C}_T$ of $\mathfrak{S}_r$ such that for any $\sigma \in \mathfrak{C}_T$ and $i = 1, \cdots, r$, $\sigma(i)$ is in the same column.

We can then define an element of $\mathbb{C}[\mathfrak{S}_r]$,

$$\begin{aligned} a_T &:= \sum_{\sigma \in \mathfrak{R}_T} \sigma, \\ b_T &:= \sum_{\sigma \in \mathfrak{C}_T} \operatorname{sgn} \sigma \sigma, \\ c_T &:= a_T b_T. \end{aligned}$$

Fact 0.1. *Let R be a ring. An element $e \in R$ is called a quasi-idempotent if there is $c \in R^\times$ such that $e^2 = ce$. Furthermore, we have the following isomorphism for any R-module M.*

$$\operatorname{Hom}_R(Re, M) \underset{[re \mapsto rem] \hookleftarrow em}{\overset{f \mapsto f(e)}{\cong}} eM.$$

Definition 0.7. *Let R be a ring and G be a finite group. For $f \in R[G]$, where*

$$f = \sum_{g \in G} f_g g,$$

we define,

$$\operatorname{tr}_{R[G]}(f) = |G| f_e.$$

Remark 0.4. *For $\sigma \in \mathfrak{S}_r$,*

$$\operatorname{tr}_{\mathbb{C}[\mathfrak{S}_r]}(\sigma) = \begin{cases} 0, & (\sigma \neq 1), \\ n!, & (\sigma = 1). \end{cases}$$

Theorem 0.3. *Let T, U be Young tableaus of shape $\lambda, \mu \vdash r$, respectively.*

1). $a_T \mathbb{C}[\mathfrak{S}_r] b_T = \mathbb{C} c_T$.

2). $\lambda > \mu \Rightarrow a_T \mathbb{C}[\mathfrak{S}_r] b_U = 0$.

3). $\mathbb{C}[\mathfrak{S}_r] c_T$ is a simple module and any simple module is isomorphic of a module of this form.

4). $\lambda = \mu \Leftrightarrow \mathbb{C}[\mathfrak{S}_r] c_T \cong \mathbb{C}[\mathfrak{S}_r] c_U$.

Proof. The first two are exercises and can be shown by direct calculation. We will show these two imply the last two. Consider the algebra anti-homomorphism,

$$\iota : \mathbb{C}[\mathfrak{S}_r] \to \mathbb{C}[\mathfrak{S}_r] = [\sigma \mapsto \sigma^{-1}].$$

Note that $\iota(a_T) = a_T, \iota(b_T) = b_T$. Thus $b_T a_T = \iota(c_T)$. Combining with 1), we obtain,

$$\mathbb{C} b_T a_t = \mathbb{C} \iota(c_T) = \iota(a_T \mathbb{C}[\mathfrak{S}_r] b_T) = b_T \mathbb{C}[\mathfrak{S}_r] a_T.$$

Also by 2), we have,

$$0 = \iota(a_T\mathbb{C}[\mathfrak{S}_r]b_U) = b_U\mathbb{C}[\mathfrak{S}_r]a_T.$$

Put them together, we have,

$$c_T\mathbb{C}[\mathfrak{S}_r]c_T = a_T(b_T\mathbb{C}[\mathfrak{S}_r]a_T)b_T \subseteq a_T\mathbb{C}[\mathfrak{S}_r]b_T = \mathbb{C}c_T.$$

Hence $c_T^2 = p(T)c_T$ for some $p(T) \in \mathbb{C}$. If $p(T) = 0$ then c_T is nilpotent. Therefore $\mathrm{tr}_{\mathbb{C}[\mathfrak{S}_r]}(c_T) = 0$. By definition,

$$c_T = a_T b_T = \sum_{\substack{\sigma\in\mathfrak{R}_T \\ \tau\in\mathfrak{C}_T}} \mathrm{sgn}(\tau)\sigma\tau.$$

As $\mathfrak{R}_T \cap \mathfrak{C}_T = \{e\}$, in particular

$$c_T = 1 + \sum_{\substack{\sigma\in\mathfrak{S}_r \\ \sigma\neq 1}} a_\sigma\sigma.$$

We obtain $\mathrm{tr}_{\mathbb{C}[\mathfrak{S}_r]} = n!$. Thus c_T is a quasi-idempotent. In particular,

$$c_T\mathbb{C}[\mathfrak{S}_r]c_T = \mathbb{C}c_T.$$

Using Fact 0.1, we obtain,

$$\mathrm{Hom}_{\mathbb{C}[\mathfrak{S}_r]}(\mathbb{C}[\mathfrak{S}_r]c_T, \mathbb{C}[\mathfrak{S}_r]c_T) \cong c_T\mathbb{C}[\mathfrak{S}_r]c_T \cong \mathbb{C}.$$

As $\mathfrak{S}_r$ is semisimple, $\mathbb{C}[\mathfrak{S}_r]c_T$ is simple.

For 4), if T and U are of the same shape then $T = \sigma U$ for some $\sigma \in \mathfrak{S}_r$. In this case, we have,

$$\mathfrak{R}_T = \mathfrak{R}_{\sigma U} = \sigma\mathfrak{R}_U\sigma^{-1}, \mathfrak{C}_T = \mathfrak{C}_{\sigma U} = \sigma\mathfrak{C}_U\sigma^{-1}.$$

In particular, $c_T = \sigma c_U\sigma^{-1}$. Using this, we see that the following map is and isomorphism.

$$\mathbb{C}[\mathfrak{S}_r] \ni m \overset{\sim}{\mapsto} m\sigma \in \mathbb{C}[\mathfrak{S}_r]c_T\sigma = \mathbb{C}[\mathfrak{S}_r]\sigma\sigma^{-1}c_T\sigma = \mathbb{C}[\mathfrak{S}_r]c_U.$$

Assume $\lambda \neq \mu$. Without loss of genearlity, we assume $\lambda > \mu$. Then

$$\mathrm{Hom}_{\mathbb{C}[\mathfrak{S}_r]}(\mathbb{C}[\mathfrak{S}_r]c_U, \mathbb{C}[\mathfrak{S}_r]c_T) \cong c_U\mathbb{C}[\mathfrak{S}_r]c_T = a_U(b_U\mathbb{C}[\mathfrak{S}_r]a_T)b_T = \{0\}.$$

□

Example 0.2. *Consider $\lambda = (2, 1)$. Then consider*

$$T = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}, \mathfrak{R}_T = \langle(13)\rangle, \mathfrak{C}_T = \langle(12)\rangle.$$

Thus we have,

$$a_T = e + (13), b_T = e - (12), c_T = e - (12) + (13) - (123).$$

Definition 0.8. *Let $\lambda \vdash r$ and pick a Young tableau T of shape λ, we define the specht module S^λ as,*

$$S^\lambda := \mathbb{C}[\mathfrak{S}_r]c_T.$$

By Theorem 0.3 4), this is independent of choice of a Young tableau up to isomorphisms.

Theorem 0.4.

$$\{S^\lambda \mid \lambda \vdash r\}$$

is the complete set of irreducible $\mathfrak{S}_r$-representations.

Notation 0.2. *Let $\lambda \vdash r$ with k-parts. We then define,*

$$\mathfrak{S}_\lambda = \prod_{i=1}^{k} \mathfrak{S}_{\lambda_i}.$$

Remark 0.5. *We have,*

$$\mathbb{C}[S_r/S_\lambda] := \mathrm{Ind}_{S_\lambda}^{S_r} \mathrm{triv} = \mathbb{C}[S_r] \underset{\mathbb{C}[S_\lambda]}{\otimes} \mathrm{triv}.$$

The isomorphism can be explicitly written as,

$$\mathbb{C}[S_r/S_\lambda] \ni [1] \leftarrow\!\shortmid\, 1 \otimes 1 \in \mathbb{C}[S_r] \underset{\mathbb{C}[S_\lambda]}{\otimes} \mathrm{triv}.$$

Theorem 0.5 (Frobenius Reciprocity)**.** *Let $\mathbb{K}$ be a field and G be a group. For a subgroup H, consider a $\mathbb{K}[H]$-module M and $\mathbb{K}[G]$-module N, we have,*

$$\mathrm{Hom}_{\mathbb{K}[G]}(\mathbb{K}[G] \underset{\mathbb{K}[H]}{\otimes} M, N) \cong \mathrm{Hom}_{\mathbb{K}[H]}(M, {}_{K[H]}N).$$

Lemma 0.4. *Let T be a Young tableau of shape λ. We have,*

$$\mathbb{C}[\mathfrak{S}_r/\mathfrak{S}_\lambda] \cong \mathbb{C}[\mathfrak{S}_r]a_T.$$

Proof. Observe that $\mathbb{R}_T \cong \mathfrak{S}_\lambda$. Therefore for any $\sigma \in \mathfrak{S}_\lambda$, we have, $\sigma a_T = a_T$. This shows $\mathbb{C}a_T \cong \mathrm{triv}$ as $\mathfrak{S}_\lambda$-representations. By Theorem 0.5, we obtain,

$$\mathrm{Ind}_{\mathfrak{S}_\lambda}^{\mathfrak{S}_r} \mathrm{triv} \ni 1 \otimes 1 \mapsto a_T \in \mathbb{C}[\mathfrak{S}_r]a_T.$$

This is surjective as a_T generates $\mathbb{C}[\mathfrak{S}_r]a_T$ as a $\mathbb{C}[\mathfrak{S}_r]$-module. We have a basis

$$\{\sigma a_T \mid \sigma \in \mathfrak{S}_r/\mathfrak{S}_\lambda\}.$$

Therefore, by counting the dimension, we conclude the morphism is an isomorphism. □

Lemma 0.5. *S^λ appears in $\mathbb{C}[\mathfrak{S}_r, \mathfrak{S}_\lambda]$ with multiplicity 1 and any other simple S^μ appearing in $\mathbb{C}[\mathfrak{S}_r/\mathfrak{S}_\lambda]$ satisfies $\mu \geq \lambda$.*

Proof. By Theorem 0.3, we have,

$$\begin{aligned}\mathrm{Hom}_{\mathbb{C}[\mathfrak{S}_r]}(\mathbb{C}[\mathfrak{S}_r]c_T, \mathbb{C}[\mathfrak{S}_r]a_T) &\cong c_T\mathbb{C}[\mathfrak{S}_r]a_T,\\ &\cong a_T(b_T\mathbb{C}[\mathfrak{S}_r]a_T),\\ &= \mathbb{C}a_Tb_Ta_T.\end{aligned}$$

Since c_T is a quasi-idempotent and $c_T = (a_Tb_T)^2$, $a_Tb_Ta_T \neq 0$. Thus

$$\dim \mathrm{Hom}_{\mathbb{C}[\mathfrak{S}_r]}(S^\lambda, \mathbb{C}[\mathfrak{S}_r/\mathfrak{S}_\lambda]) = 1.$$

We have proven the first part. For the second part,

$$\mathrm{Hom}_{\mathbb{C}[\mathfrak{S}_r]}(\mathbb{C}[\mathfrak{S}_r]c_U, a_T) \cong c_U\mathbb{C}[\mathfrak{S}_r]a_Ta_U = \underbrace{(b_U\mathbb{C}[\mathfrak{S}_r]a_T)}_{\substack{=0 \text{ if } \mu<\lambda\\ \text{by Theorem 0.3}}}.$$

□

Definition 0.9.

$$\mathcal{O}(\mathfrak{S}_r)^{\mathfrak{S}_r} = \mathrm{Map}(\{\textit{conjugacy classes of } \mathfrak{S}_r\}, \mathbb{C}).$$

For $f \in \mathcal{O}(\mathfrak{S}_r)^{\mathfrak{S}_r}$ and $\lambda \vdash r$, we set,

$$f(\lambda) = f(\sigma),$$

for any $\sigma \in \mathfrak{S}_r$ of cycle type λ.

Proposition 0.3. *Let G be a finite group and H be a subgroup of it. The character of permutation representation $\mathbb{C}[G/H]$ is*

$$\chi_{\mathbb{C}[G/H]}(g) = \sum_{i\in I}\frac{|C_G(g)|}{|C_H(h_i)|}.$$

where $\{h_i\}_{i\in I}$ is the complete set of representatives of the H-conjugacy classes in

$$H \cap (\{xgx^{-1} \mid x \in G\}).$$

More generally, for any H-representation V, we have,

$$\chi_{\mathrm{Ind}_H^G(V)}(g) = \sum_{i\in I}|C_G(g)||C_H(h_i)\chi_V(h_i).$$

Proof. Let $g_1, \cdots, g_m \in G$ be a complete list of coset representations G/H. Let $v_1, \cdots, v_n$ be a basis of V, then

$$\{g_i \otimes v_j\}_{\substack{1\le i\le m\\ 1\le j\le n}}$$

is a basis of $\mathrm{Ind}_H^G(V)$. For a vectorspace W and a linear map $\varphi : W \to W$. Pick a basis $\{w_1, \cdots, w_k\}$, we have

$$\mathrm{tr}(\varphi) = \sum_{i=1}^{k}\text{coefficient of } w_i \text{ in } \varphi(w_i).$$

Let $g \in G$ then $gg_i = g_l h$ for some l and $h \in H$, thus,

$$g(g_i \otimes v_j) = g_l \otimes hv_j.$$

If $i \neq l$ then the coefficient of $g_i \otimes v_j$ in $g_l \otimes hv_j$ is 0. Thus this has no contribution to the equality,

$$\chi_{\mathrm{Ind}_H^G(V)}(g) = \mathrm{tr}_{\mathrm{Ind}_H^G(V)}(g).$$

If $i = l$ then $g_i \otimes V$ is stable under g-action. Note that the trace of g on $g_i \otimes V$ is the trace of $h = g_i^{-1} g g_i$ on V hence,

$$\chi_{\mathrm{Ind}_H^G(V)}(g) = \sum_{xH \in (G/H)^g} \chi_V(x^{-1}gx) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \chi_V(x^{-1}gx).$$

Let

$$X_g := \{x \in G \mid x^{-1}gx \in H\}.$$

Then there is a surjective map,

$$X_g \ni x \overset{\pi}{\longmapsto} x^{-1}gx \in H \cap \{x^{-1}xg \mid x \in G\}.$$

Note that

$$x^{-1}gx = y^{-1}gy \Rightarrow (xy^{-1})g(xy^{-1}) = g.$$

Therefore each fiber of π is of size $|C_G(g)$. In particular,

$$|\pi^{-1}(\{H\text{-conjugacy class of } h_i\}) = |C_G(g)| \frac{|H|}{|C_H(h_i)|}.$$

Using this result, we hae,

$$\begin{aligned}
\chi_{\mathrm{Ind}_H^G(V)}(g) &= \frac{1}{|H|} \sum_{x \in X_g} \chi_V(x^{-1}gx), \\
&= \frac{1}{|H|} \sum_{x \in X_g} |C_G(g)| \frac{|H|}{|C_H(h_i)|} \chi_V(h_i), \\
&= \sum_{i \in I} \frac{|C_G(g)|}{|G_H(h_i)|} \chi_V(h_i).
\end{aligned}$$

□